Section A

1 **(a)** angle (subtended) at centre of circle B1
(by) arc equal in length to radius B1 [2]

(b) **(i)** point S shown below C B1 [1]

(ii) (max) force / tension = weight + centripetal force C1
centripetal force = $mr\omega^2$ C1
$15 = 3.0/9.8 \times 0.85 \times \omega^2$ C1
$\omega = 7.6 \text{ rad s}^{-1}$ A1 [4]

2 **(a)** **(i)** 27.2 + 273.15 or 27.2 + 273.2 C1